E-Commerce Electronics Sales Report

1. Project Overview

This project analyses an e-commerce electronics sales dataset containing transactional records for the entire year of 2019. The dataset consists of monthly sales data across various electronic products such as phones, laptops, monitors, and accessories.

The objective of this project is to identify key sales patterns, product performance, purchasing behaviour, and geographic trends to derive business insights.

The analysis pipeline combines **Python for data preprocessing** and **Power BI for interactive visualization**, transforming raw transactional data into an analytical dashboard that supports data-driven decision making.

2. Dataset Summary

Dataset Structure:

Total records:

500,000+

Columns:

- Order ID
- Product
- Quantity Ordered
- Price Each
- Order Date
- Purchase Address

3. Data Cleaning & Preparation (Python)

Data preprocessing was performed using **Python (Pandas)**.

Key Data Preparation Steps

• Removing empty rows and corrupted records

• Converting numeric columns to proper data types
• Parsing the order date column into datetime format
• Creating new analytical features:

Derived columns:

- sales = quantity_ordered * price_each
- month
- hour
- city
- state
- week_type (weekday / weekend)
- product_category

Product Categories Created:

Products were grouped into broader categories for better analysis:

- Phones
- Laptops
- Monitors
- Charging Accessories
- Batteries
- TV
- Other

These transformations improved data quality and enabled more meaningful analytics.

4. Exploratory Data Analysis

Several analytical questions were explored to understand sales behavior:

- What is the total revenue generated in 2019?
- Which product categories generate the highest revenue?
- Which cities produce the highest sales?
- What time of day do customers place the most orders?
- How do sales differ between weekdays and weekends?
- What are the monthly sales trends across the year?
- Which product categories contribute most to revenue?

5. Power BI Dashboard

An **interactive E-Commerce Sales Dashboard** was built using Power BI.

The dashboard presents key business metrics and insights through interactive visualizations.

Key Metrics:

• Total Revenue
• Total Orders
• Total Quantity Sold
• Average Order Value

Visual Insights:

• Revenue by product category
• Sales distribution by city
• Monthly sales trend
• Orders by hour of day
• Weekday vs weekend sales comparison

Interactive Filters:

The dashboard includes slicers for:

Product Category
City

These filters allow users to dynamically explore sales trends and product performance.

6. Key Insights

Important insights derived from the analysis include:

• Total revenue for 2019 exceeded **$103 million**.

• **Phones and laptops** generate the highest share of product revenue.

• **San Francisco and Los Angeles** are the top performing cities in terms of sales.

• Customer purchases peak during:

11 AM – 1 PM
6 PM – 8 PM

suggesting optimal advertisement timing.

• Sales activity is **higher on weekdays compared to weekends**, indicating strong workday purchasing patterns.

• Sales gradually increase throughout the year, with **December showing the strongest performance**, likely due to holiday shopping.